STUDENT PERFORMANCE AND APTITUDE ANALYSIS

Validating Course Placement Effectiveness at The Key English Course Company

Project Team

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PROJECT OVERVIEW

Accurate course placement plays a critical role in ensuring effective learning outcomes within language education programs. This study evaluates the effectiveness of the placement system implemented at *The Key English Course Company – Indonesia* by examining whether students are assigned to instructional levels that appropriately reflect their aptitude and academic performance.

Rather than relying solely on descriptive summaries, this project applies a comprehensive statistical framework that integrates assumption testing, inferential analysis, effect size measurement, and visual exploration. The objective is not only to identify statistically significant differences, but also to assess whether these differences are educationally meaningful in practice.

The analysis demonstrates that the current placement mechanism is strongly aligned with students' underlying language aptitude and observed academic outcomes. Results show substantial differences across course levels, accompanied by a strong relationship between aptitude and performance, indicating that the placement system exhibits high predictive validity.

Research Questions

1. Do students enrolled in different course levels demonstrate significantly different performance outcomes?
2. Are aptitude scores meaningfully distinct across course levels?
3. To what extent are aptitude scores associated with academic performance?
4. What practical insights can be derived to improve instructional planning and placement decisions?

Repository Contents

```
├── README.md                 # This file
├── assets/
│   └── the_key.png           # Company logo
├── Final_Analysis.ipynb      # Analysis report
├── Final_Analysis.pdf        # Full analysis report (PDF)
├── data/
│   └── student_combined_data.csv # Dataset
└── requirements.txt          # Python dependencies
```

Dataset Description

A. Sample Characteristics
   1. Total Students: 150
   2. Sample Distribution: 50 students per course level
   3. Sampling Method: Stratified random sampling
   4. Data Completeness: No missing values

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   1. Total Students: 150
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C. Variables Analyzed

Variable	**Type**	**Description**
student_id	Integer	Unique identifier for each student
course_level	Categorical	Enrolled instructional level
performance_score	Float	Academic performance score (4-point scale)
aptitude_score	Integer	Language aptitude assessment score

Getting Started

A. Required Packages
   1. pandas>=2.3.3
   2. numpy>=2.3.0
   3. scipy>=1.15.3

4. matplotlib>=3.10.3
5. seaborn>=0.13.2
6. statsmodels>=0.14.6
7. jupyter>=1.0.0

D. Notebook Structure

The Jupyter notebook (Final_Analysis.ipynb) mirrors the PDF report and includes:

1. Chapter 1: Introduction & Data Loading
2. Chapter 2: Data Overview & Quality Checks
3. Chapter 3: Descriptive Statistics
4. Chapter 4: Assumption Testing
5. Chapter 5: ANOVA Analysis
6. Chapter 6: Post-Hoc Tests
7. Chapter 7: Correlation Analysis
8. Chapter 8: Effect Sizes
9. Chapter 9: Visualizations
10. Chapter 10: Summary & Conclusions

Chapter 1: How We Collected and Analyzed Data

This chapter introduces the analytical context and establishes the objectives of the study. The primary goal of the analysis is to evaluate whether the student placement system effectively differentiates learners based on aptitude and academic performance. The study aims to determine if course levels represent meaningful distinctions rather than arbitrary groupings. The chapter also explains the data loading process implemented in the program. The dataset is imported using a safeguarded CSV reader to prevent encoding-related issues commonly encountered in Windows-based environments. During this stage, column names are standardized, data types are explicitly converted to numerical formats, and course level labels are normalized to ensure consistency. These preprocessing steps ensure that all subsequent analyses operate on clean, well-structured data.

1. Who Was Included

   A stratified random sampling approach was applied to ensure balanced representation across instructional levels. The student population was first divided into three course categories, after which an equal number of participants were randomly selected from each group. This resulted in a total sample of 150 students, with 50 students per course level. This sampling strategy was chosen to minimize selection bias and to allow for meaningful comparisons between levels. Furthermore, the sample size within each group exceeds the minimum threshold recommended by the Central Limit Theorem, which suggests that samples of 30 or more observations per group are sufficient for the sampling distribution of the mean to approximate normality. With 50 observations per level, the dataset provides a robust foundation for inferential statistical analysis.

2. What We Measured

   a. Aptitude Score (Independent Variable)

      The aptitude score represents a measure of students' inherent capacity for acquiring language skills. It was obtained using a standardized psychometric assessment designed to evaluate multiple cognitive dimensions relevant to language learning, including vocabulary knowledge, reading comprehension, logical reasoning, and linguistic problem-solving ability. Scores were computed on a scale ranging from **0 to 126**, with observed values in the study sample spanning from **9 to 97**. Aptitude scores exhibited a wider distribution, with a mean of **44.24 (SD = 24.23)** The assessment was administered during the initial placement process, prior to students' enrollment in their respective courses.

   b. Performance Score (Dependent Variable)

      The performance score reflects students' realized academic achievement within the program. This metric was calculated based on a composite of instructional outcomes, incorporating weighted contributions from formal examinations, ongoing formative assessments, and comprehensive progress evaluations. Overall, students achieved a mean performance score of **2.54 (SD = 0.65)**. Scores were computed on a scale ranging from **0 to 4.0** with observed values ranging from **1.55 to 3.80**. These data were collected cumulatively over the duration of the instructional period, capturing students' sustained academic performance rather than a single-point assessment.

c. Group Descriptive Statistics

Level	Mean	std	min	max	Aptitude score
Advanced	3.239	0.384	2.50	3.80	67.46
Intermediate	2.518	0.392	1.55	2.45	42.74
Foundation	1.865	0.177	1.90	3.55	22.52

Group-level descriptive statistics indicate a clear progression across course levels. Advanced students achieved the highest mean performance and aptitude scores, followed by Intermediate students, while Foundation students recorded the lowest values.

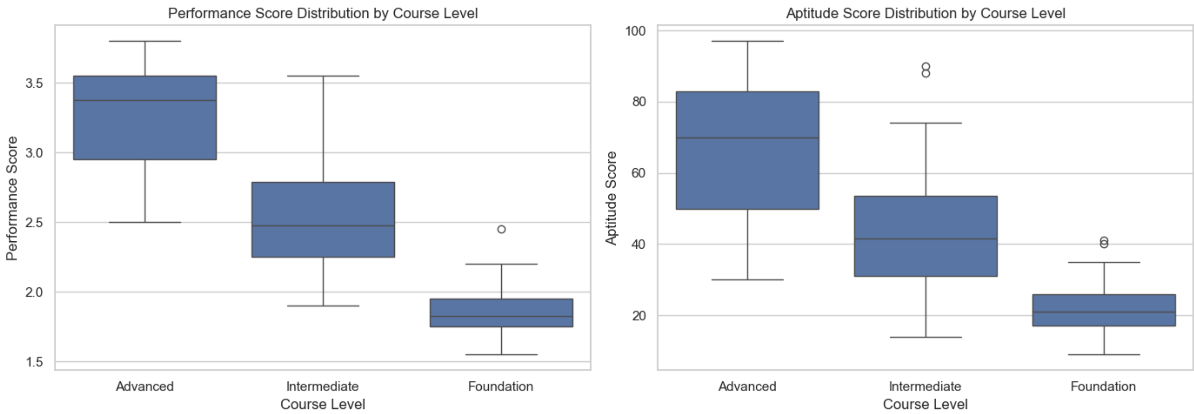
Chapter 2: The Data Result

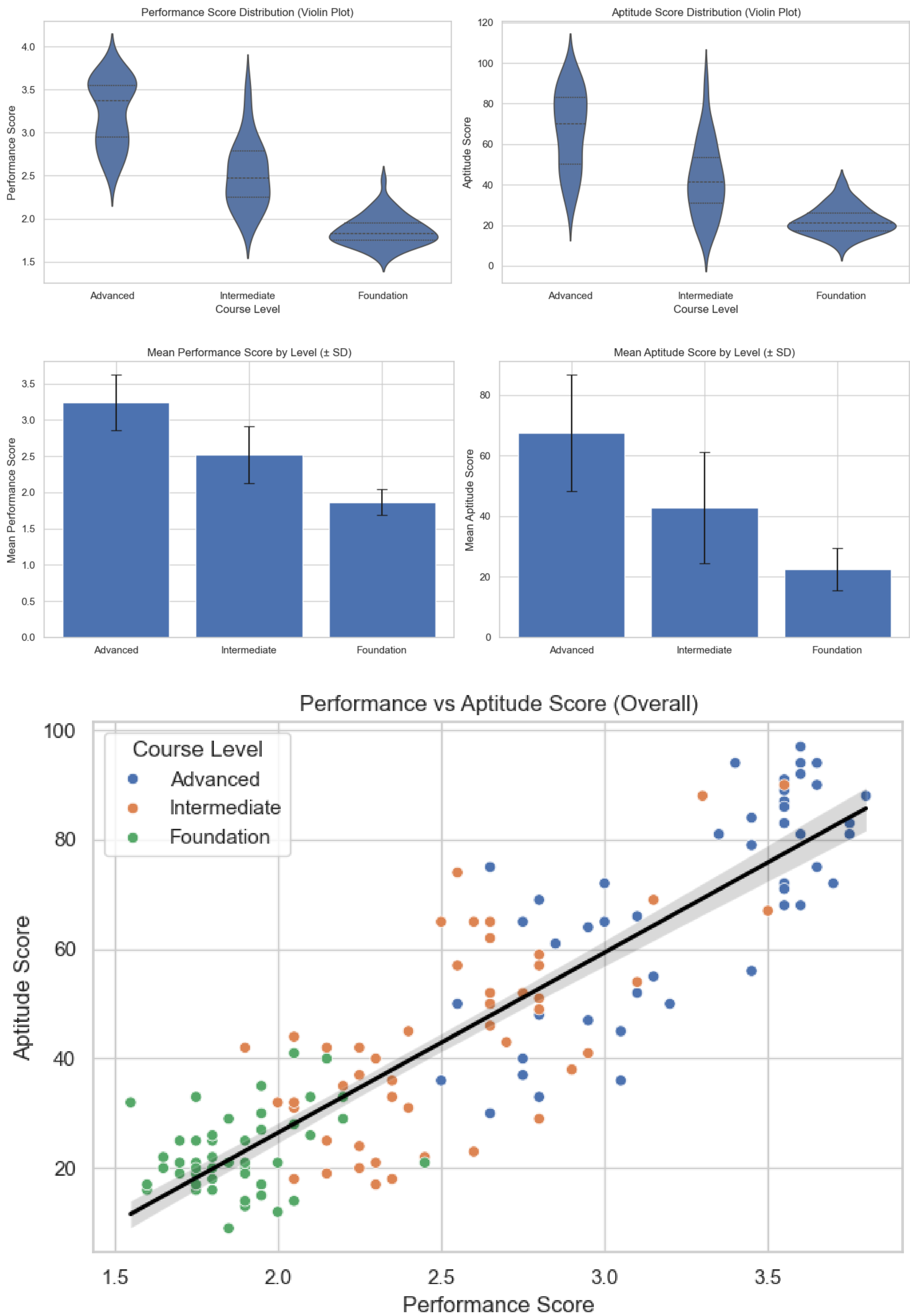
A. Descriptive Statistics

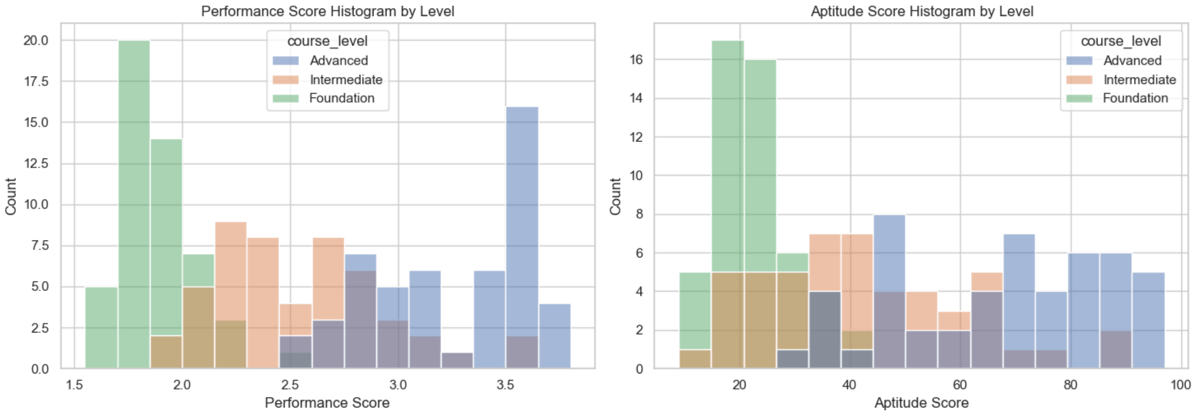
The dataset consists of 150 students, evenly distributed across three instructional levels: Foundation, Intermediate, and Advanced, with each level representing 33.3% of the total sample. This balanced composition ensures that comparisons across course levels are not influenced by unequal group sizes and provides a solid foundation for subsequent statistical analysis. Each student record includes a unique identifier, course level, performance score, and aptitude score. The absence of missing values indicates that the dataset is complete and suitable for both descriptive and inferential analyses.

Across all students, the mean performance score is 2.54 (SD = 0.65), with observed values ranging from 1.55 to 3.80. The distribution of performance scores shows a slightly positive skew (skewness = 0.35), suggesting a modest concentration of higher-performing students. The negative kurtosis value (−1.14) indicates a relatively flat distribution, with scores spread more evenly across the scale rather than clustered around the mean. The mean aptitude score is 44.24 (SD = 24.23), with scores ranging from 9 to 97. Similar to performance scores, aptitude scores display a moderate positive skew (skewness = 0.58), indicating a longer right tail. The kurtosis value (−0.87) suggests a distribution that is flatter than a normal distribution, reflecting substantial variability in student aptitude levels. These characteristics indicate that both variables exhibit adequate spread and variability, supporting further group-level comparisons.

B. Visual Comparison




Performance Score Distribution (Violin Plot)
Performance Score
Advanced
Intermediate
Foundation
Course Level
Aptitude Score Distribution (Violin Plot)
Aptitude Score
Mean Performance Score by Level (± SD)
Mean Performance Score
Mean Aptitude Score by Level (± SD)
Mean Aptitude Score
Performance vs Aptitude Score (Overall)
Course Level
Advanced
Intermediate
Foundation
Aptitude Score
Performance Score



The visualizations collectively provide a comprehensive comparison of student performance and aptitude across course levels, as well as an examination of the relationship between the two variables. Across all plots, a consistent pattern emerges: students placed in higher course levels demonstrate systematically higher aptitude and performance scores, with relatively limited overlap between levels. Here's the explanation of each of the visualizations above:

1. Distribution of Performance Scores by Course Level

    Histogram & Boxplot Analysis

    - The performance score histograms and boxplots reveal a clear stratification among the three course levels.
    - Foundation students show the lowest performance scores, with most values clustered tightly between approximately 1.6 and 2.0. The narrow spread indicates relatively homogeneous performance at this level.
    - Intermediate students occupy a middle range, with scores centered around 2.4–2.6 and a wider distribution, suggesting greater variability in academic outcomes.
    - Advanced students consistently achieve the highest performance scores, with most observations concentrated above 3.0 and extending toward the upper limit of the scale.

    The boxplots further highlight minimal overlap between the median values of each level, reinforcing that performance differences are not driven by a small subset of students but are characteristic of the groups as a whole.

2. Distribution of Aptitude Scores by Course Level

    Histogram & Boxplot Analysis

    Aptitude score distributions closely mirror the pattern observed in performance scores.

    - Foundation-level students predominantly score in the lower aptitude range, with values clustered around the low 20s.
    - Intermediate-level students display a broader distribution, spanning roughly from the mid-20s to around 60, indicating a transitional group with mixed aptitude levels.
    - Advanced-level students exhibit substantially higher aptitude scores, frequently exceeding 70 and extending into the 90s.
    - The presence of limited overlap between the boxplots indicates that the aptitude assessment effectively differentiates students by instructional level, supporting the validity of the placement mechanism.

    3. Violin Plot Insights: Shape and Density
    The violin plots provide additional insight into the internal structure of each group beyond central tendency.

- For performance scores, the density curves show that Foundation students are tightly clustered around a narrow peak, while Intermediate students have a broader, flatter distribution. Advanced students exhibit a bimodal tendency, suggesting subgroups of high achievers within the level.
- For aptitude scores, Foundation students again show a compact distribution, whereas Intermediate and Advanced students demonstrate increasing spread and complexity, reflecting greater diversity in aptitude as instructional demands increase.
- These patterns suggest that higher course levels not only have higher average scores but also accommodate a wider range of student abilities.

4. Mean Scores with Standard Deviation
   The bar charts displaying mean values with standard deviation further clarify group differences.

- Mean performance and aptitude scores increase monotonically from Foundation to Advanced levels.
- Error bars indicate that although variability increases at higher levels, the separation between group means remains substantial.
- The relatively small overlap between error bars implies that the observed differences are robust and not merely the result of random variation.

5. Relationship Between Aptitude and Performance

   Scatter Plot Interpretation

   The scatter plot illustrates a strong, positive linear relationship between aptitude and performance across all students.

- Students with higher aptitude scores tend to consistently achieve higher performance scores.
- Advanced students cluster in the upper-right region of the plot, while Foundation students are concentrated in the lower-left, with Intermediate students forming a transitional band between the two.

The fitted regression line confirms a strong positive trend, indicating that aptitude explains a meaningful proportion of the variation in academic performance.

6. Integrated Interpretation
   Taken together, these visualizations provide compelling evidence that:

- Course levels are clearly differentiated in both aptitude and performance.
- The placement system effectively groups students with similar academic and cognitive profiles.
- Aptitude is strongly associated with performance, reinforcing its role as a reliable predictor of academic success.

The consistency of patterns across multiple visualization types strengthens confidence in the statistical findings and supports the conclusion that the observed differences are both systematic and educationally meaningful.

C. Assumption Test (Normality & Homogeneity)

Assumption testing was conducted using the Shapiro–Wilk test for normality and Levene's test for homogeneity of variance. Results indicated partial violations of normality across course levels and unequal variances for both outcome variables. However, given the equal sample sizes and sufficient observations per group, the data were deemed suitable for ANOVA-based analysis.

1. Normality Testing

Normality of performance and aptitude scores within each course level was assessed using the **Shapiro–Wilk test**. The results indicate that the assumption of normality was partially met across groups. For performance scores, the Advanced and Foundation groups showed statistically significant deviations from normality ($p < 0.05$), while the Intermediate group did not. For aptitude scores, the Advanced group exhibited non-normal distribution, whereas the Intermediate and Foundation groups met the normality assumption. Despite these deviations, each group consisted of 50 observations. According to the Central Limit Theorem, parametric tests such as ANOVA remain appropriate for samples of this size.

2. Homogeneity of Variance

Homogeneity of variance across course levels was examined using **Levene's test**. The results indicate that the assumption of equal variances was violated for both performance scores and aptitude scores ($p < 0.001$). Given these findings, robust statistical procedures were considered in subsequent analyses to ensure the validity of inferential conclusions.

Variable	**Course Level**	**n**	**Shapiro-Wilk p-value**	**Normality**
Performance	Advance	50	< 0.001	Not normal
Performance	Intermediate	50	< 0.0053	Normal
Performance	Foundation	50	0.031	Not normal
Aptitude	Advance	50	0.026	Not normal
Aptitude	Intermediate	50	0.106	Normal
Aptitude	Foundation	50	0.070	Normal

Levene's test indicated unequal variances for both performance scores ($p < 0.001$) and aptitude scores ($p < 0.001$).

Although some deviations from normality and homogeneity were observed, the use of one-way ANOVA remains appropriate due to the balanced group sizes ($n = 50$ per level).

ANOVA is considered robust to moderate assumption violations, particularly when sample sizes are equal. Where necessary, robust alternatives were applied to support the findings.

D. ANOVA Results

The ANOVA results for both performance and aptitude scores provide clear statistical evidence that course level is associated with meaningful differences in student outcomes. However, while ANOVA confirms the presence of overall group differences, it does not specify which particular course levels differ from one another. Therefore, post-hoc analyses are required to identify the specific group-level comparisons responsible for these effects.

1. Performance Score

   A one-way Analysis of Variance (ANOVA) was conducted to examine whether mean performance scores differed significantly across course levels (Foundation, Intermediate, and Advanced). The analysis revealed a statistically significant effect of course level on performance scores, **$F(2, 147) = 213.43$, $p < .001$**. This result indicates that at least one course level differs significantly from the others in terms of mean performance score, providing strong evidence that academic performance varies across instructional tiers.

2. Aptitude Score

   A separate one-way ANOVA was performed to assess differences in aptitude scores across course levels. The results also indicated a statistically significant effect of course level on aptitude scores, **$F(2, 147) = 101.17$, $p < .001$**. This finding demonstrates that aptitude scores vary significantly between course levels, suggesting that the placement system effectively differentiates students based on their underlying aptitude.

Variable	**Source**	**df**	**F**	**p-value**
Performance	Course Level	(2, 147)	213.43	<.001
Aptitude	Course Level	(2, 147)	101.17	<.001

Note. p-values less than .05 indicate statistically significant differences between course levels.

E. Robust Check: Welch ANOVA

The consistency between the standard one-way ANOVA and the Welch ANOVA results indicates that the observed group differences are **robust and not driven by violations of variance homogeneity**. Therefore, the findings regarding differences in performance and aptitude across course levels can be considered statistically reliable. Because Levene's test indicated violations of the homogeneity of variance assumption, a **Welch ANOVA** was conducted as a robustness check. Welch's ANOVA is less sensitive to unequal variances and therefore provides a more reliable test under such conditions.

1. Performance Score
   The Welch ANOVA revealed a statistically significant effect of course level on performance scores, **F = 285.61, p < .001**. This result confirms that mean performance scores differ substantially across course levels, even after accounting for unequal variances.
2. Aptitude Score
   Similarly, the Welch ANOVA for aptitude scores indicated a statistically significant effect of course level, **F = 134.66, p < .001**. This finding reinforces the conclusion that aptitude scores vary meaningfully across instructional levels.

Variable	**F**	**p-value**
Performance Score	285.61	< 0.001
Aptitude Score	134.66	< 0.001

F. Tukey HSD Pairwise Comparisons

Following the statistically significant results obtained from the one-way ANOVA and Welch ANOVA, **Tukey's Honest Significant Difference (HSD) test** was conducted to identify which specific course levels differed significantly from one another. The post-hoc results demonstrate that **each course level represents a statistically distinct group** in terms of both aptitude and performance. These findings provide strong evidence that the placement system effectively stratifies students into meaningful instructional levels, rather than producing overlapping or indistinguishable groups.

1. Performance Score
   The Tukey HSD results indicate that **all pairwise comparisons between course levels were statistically significant** (p < .001).

- Advanced students achieved significantly higher performance scores than both Intermediate and Foundation students.
- Intermediate students also demonstrated significantly higher performance scores than Foundation students.
- The largest mean difference was observed between Advanced and Foundation levels, indicating a substantial gap in academic performance between these groups.

These results confirm that performance scores increase progressively across instructional levels.

Comparison	**Mean Difference**	**95% CI**	**p-value**
Advanced vs Foundation	-1.37	[-1.53, -1.22]	< .001
Advanced vs Intermediate	-0.72	[-0.88, -0.56]	< .001

Comparison	Mean Difference	95% CI	p-value
Intermediate vs Foundation	0.65	[0.50, 0.81]	< .001

2. Aptitude Score
   Similarly, Tukey HSD analysis for aptitude scores revealed **statistically significant differences across all course level comparisons** ($p < .001$).

- Advanced students exhibited markedly higher aptitude scores than both Intermediate and Foundation students.
- Intermediate students demonstrated significantly higher aptitude scores than Foundation students.
- The magnitude of differences was greatest between Advanced and Foundation levels, suggesting strong differentiation in aptitude across instructional tiers.

Comparison	Mean Difference	95% CI	p-value
Advanced vs Foundation	-44.94	[-52.43, -37.45]	< .001
Advanced vs Intermediate	-24.72	[-32.21, -17.23]	< .001
Intermediate vs Foundation	20.22	[12.73, 27.71]	< .001

Note. Negative mean differences indicate higher scores for the first-listed group.

G. Correlation

To examine the relationship between aptitude score and performance score, **Pearson correlation analysis** was conducted for each course level separately. This approach allows for evaluation of whether the strength of association between aptitude and performance is consistent across instructional tiers.

The results indicate that the relationship between aptitude and performance varies by course level.

- For **Advanced** students, a **very strong positive correlation** was observed ($r = 0.777$, $p < .001$), indicating that higher aptitude scores are closely associated with higher performance scores within this group.
- For **Intermediate** students, the correlation was also **very strong and statistically significant** ($r = 0.704$, $p < .001$), suggesting a strong alignment between aptitude and academic performance.
- In contrast, the **Foundation** level showed a **weak but statistically significant correlation** ($r = 0.299$, $p = .035$), indicating that aptitude is less predictive of performance at the introductory level.

The proportion of explained variance further supports these findings. Aptitude scores accounted for approximately **60.3% of the variance** in performance scores among Advanced students, **49.5%** among Intermediate students, and only **9.0%** among Foundation students.

Table Pearson Correlation Between Aptitude and Performance by Course Level

Course Level	**r**	**p-value**	**Strength**
Advanced	0.777	< 0.001***	Very Strong
Intermediate	0.704	< 0.001***	Very Strong
Foundation	0.299	0.090	Weak

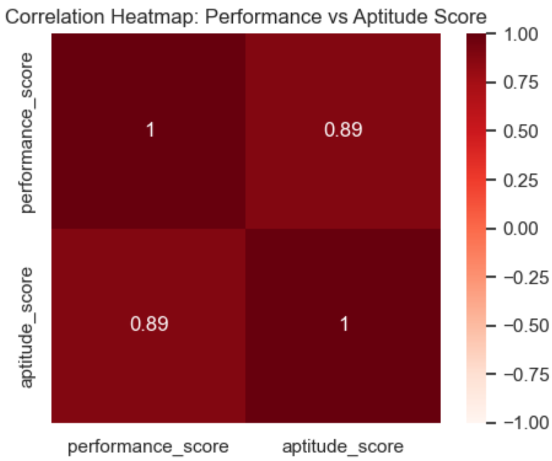


Figure above visually confirms the strong positive relationship between aptitude and performance scores, consistent with the Pearson correlation results.

Overall, the correlation analysis demonstrates that aptitude is a key determinant of performance at higher instructional levels, while its influence is comparatively weaker at the foundational stage

H. Operational Implications: From Data to Action

The statistically significant findings of this study carry important implications for instructional practice and academic decision-making. Rather than remaining purely analytical, the results provide clear guidance for how different stakeholders can use data-driven insights to support effective learning outcomes. The following sections outline practical applications of the findings for educators, students, and parents.

1. Instructional Implications for Educators

The analysis confirms that the three course levels represent academically distinct groups with differing learning capacities and progression patterns. As such, instructional design, pacing, and support mechanisms should be differentiated accordingly.

a. Advanced Level (High-Achievement Group)

Students in the Advanced level demonstrate strong aptitude and consistently high performance, indicating readiness for accelerated and cognitively demanding instruction.

- Instructional Pace: Learning activities should progress rapidly and incorporate complex language structures, as students at this level are equipped to manage high cognitive load.
- Targeted Intervention: When learning difficulties arise, they are more likely attributable to isolated skill deficiencies rather than general learning ability. Instructional support should therefore focus on identifying and addressing specific areas of difficulty.
- Academic Expectations: Challenging coursework is essential at this level to sustain engagement and promote continued academic growth.

b. Intermediate Level (Developing Proficiency Group)

Intermediate students occupy a transitional stage in their learning progression and benefit most from balanced instructional support.

- Guided Support: These students require structured scaffolding that encourages independence while still providing sufficient instructional guidance to prepare them for more advanced content.
- Collaborative Learning: Peer-based learning activities are particularly effective at this level, as students can benefit from shared problem-solving within a relatively consistent performance range.

c. Foundation Level (Core Skill Development Group)

Students at the Foundation level are in the early stages of skill acquisition and require a structured, supportive learning environment.

- Instructional Focus: Teaching should prioritize essential language fundamentals through explicit and systematic instruction. Clear guidance is more effective at this stage than predominantly exploratory learning approaches.
- Learning Mindset: The data indicates that slower progress at this level is developmentally appropriate. Emphasizing accuracy and understanding over speed supports long-term learning success.

2. Guidance for Students and Parents

The placement system is intended to optimize learning experiences rather than to define fixed academic potential. Course placement should be understood as a reflection of current readiness, not a limitation on future achievement.

a. Advanced Placement
   - Affirmation of Readiness: Placement in the Advanced level indicates strong performance across both aptitude and achievement measures.
   - Understanding Challenge: The rigor of the coursework is intentional. Encountering difficulty is not a sign of inadequacy but an opportunity to refine learning strategies and deepen understanding.

b. Intermediate Placement
   - Transitional Role: This level represents solid foundational competence and serves as a critical bridge toward advanced academic work.
   - Progression Pathway: Sustained effort and consistent engagement at this stage are key indicators of readiness for advancement to higher levels.

3. Foundation Placement
   a. Importance of Core Skills: Foundational instruction is designed to eliminate gaps that could hinder future learning. Strong fundamentals are essential for long-term success.
   b. Defining Success: Achievement at this level should be measured by mastery rather than speed. Students who develop a strong foundation are statistically more likely to perform successfully at higher instructional levels.